Eco-Track: Smart Waste Management System Using IoT and Route Optimization

Submitted by,

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Abstract

Eco-Track is an intelligent and technology-driven waste management system designed to enhance urban cleanliness and environmental sustainability through the integration of Internet of Things (IoT) technologies. The system utilizes IoT-enabled smart bins equipped with sensors that continuously monitor garbage levels in real time. These sensors detect the fill status of waste containers and transmit data to a centralized monitoring platform, enabling authorities to track waste accumulation efficiently and prevent overflow situations.

In addition to real-time monitoring, Eco-Track incorporates a citizen participation feature that allows individuals to report instances of illegal dumping through a user-friendly interface. This promotes community involvement and ensures quicker identification and resolution of sanitation issues. The collected data is further processed using route optimization algorithms, which generate efficient collection paths for municipal waste management vehicles. This reduces fuel consumption, operational costs, and response time while improving overall service efficiency.

By combining smart sensing, data analytics, and community engagement, Eco-Track aims to create a cleaner, more organized, and environmentally responsible waste management ecosystem for modern cities.

2. Introduction

2.1 Importance of Waste Management

Effective waste management plays a vital role in maintaining public health, environmental sustainability, and overall quality of life in urban and rural areas. With the rapid increase in population and consumerism, the volume of solid waste generated daily has significantly increased. Improper disposal of waste leads to serious environmental issues such as air, water, and soil pollution. It also contributes to the spread of diseases, unpleasant living conditions, and ecological imbalance. A well-organized waste management system ensures systematic collection, segregation, transportation, and disposal of waste, thereby reducing environmental impact and promoting a cleaner and healthier society.

2.2 Urban Challenges

Urban areas face unique and complex challenges in managing waste effectively. Rapid urbanization, industrial growth, and changing lifestyles have led to increased waste generation,

often exceeding the capacity of traditional waste collection systems. Overflowing bins, irregular waste collection schedules, traffic congestion affecting garbage trucks, and illegal dumping are common issues in cities. Additionally, limited manpower, inefficient route planning, and lack of real-time monitoring further reduce operational efficiency. These challenges highlight the limitations of conventional waste management approaches and the urgent need for modernization.

2.3 Need for a Smart System

To address these growing challenges, there is a clear need for a smart and technology-driven waste management system. A smart system leverages Internet of Things (IoT), sensors, data analytics, and automation to monitor waste levels in real time and optimize collection processes. By using intelligent bins and centralized monitoring platforms, authorities can track fill levels, prevent overflow, and deploy waste collection vehicles only when necessary. Smart route optimization reduces fuel consumption, operational costs, and carbon emissions. Furthermore, integrating citizen participation features encourages community involvement and improves accountability. Such a system enhances efficiency, transparency, and sustainability in urban waste management.

2.4 SDG 11 Alignment

The Eco-Track system aligns closely with **Sustainable Development Goal 11 (SDG 11): Sustainable Cities and Communities**, established by the United Nations. SDG 11 focuses on making cities inclusive, safe, resilient, and sustainable. Proper waste management is a critical component of this goal, as it directly impacts environmental protection, public health, and urban sustainability. By promoting cleaner surroundings, reducing pollution, optimizing resource usage, and encouraging citizen engagement, a smart waste management system contributes to building smarter and more sustainable cities. Therefore, implementing Eco-Track supports global sustainability efforts while addressing local urban challenges effectively.

3. Problem Statement

Urban areas lack real-time monitoring and optimized waste collection, leading to inefficiency and environmental hazards. The lack of immediate communication channels between citizens and waste management bodies reduces system efficiency.

4. Objectives

- Monitor bin fill levels using IoT.
- Allow citizens to report issues.
- Provide centralized dashboard.
- Optimize garbage truck routes.
- Reduce fuel and operational cost.

5. SYSTEM ARCHITECTURE DESIGN

Main Components

1 Smart Bin Unit

- Ultrasonic sensor.
- ESP32.
- Wi-Fi module.
- Power supply.

2 Citizen Mobile App

- Report waste.
- Upload photo.
- Location tracking.

3 Cloud Server

- Stores data.
- Runs route optimization.
- Updates status.

4 Admin Dashboard

- View bin status.
- View complaints.

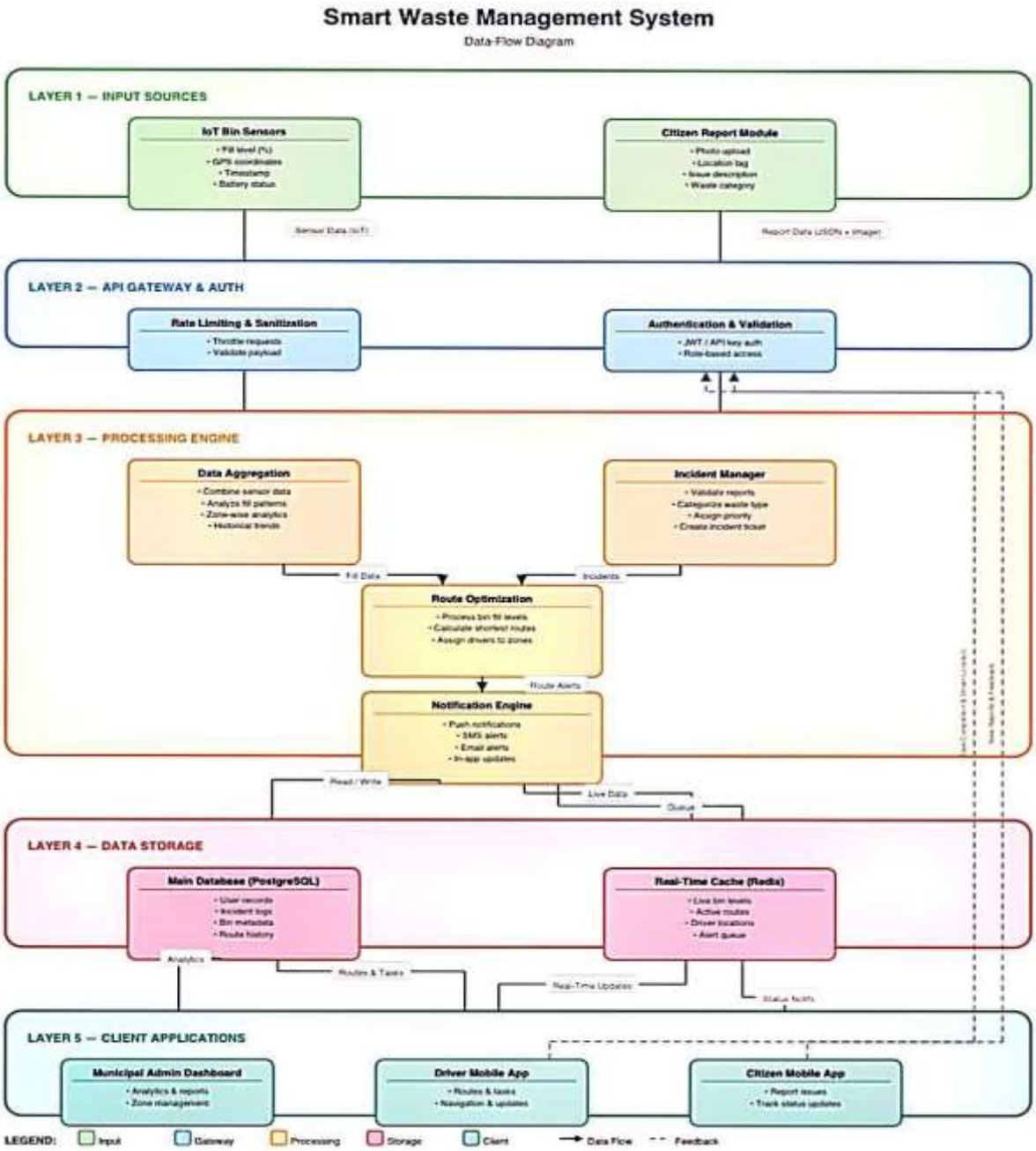
- Assign drivers.

5 Driver Mobile App

- View optimized route.
- Update collection status.

6. DATA FLOW

The Smart Waste Management System is designed to efficiently manage waste collection through continuous data monitoring, real-time incident reporting, and automated route optimization. The data flow begins with input sources such as loT-enabled smart bin sensors and citizen mobile applications. Smart bins equipped with ultrasonic sensors continuously measure the fill level of waste containers and transmit bin status information such as capacity level, location coordinates, and battery condition through an lot gateway. Simultaneously, citizens can report waste-related issues including bin overflow, illegal dumping, or damaged bins using a mobile application that captures location data, category of waste, and optional images. All incoming data is routed through an API gateway where authentication and validation mechanisms ensure secure access. Rate limiting and payload sanitization protect the system against invalid or malicious inputs. Once validated, the processing engine aggregates bin fill data and incident reports to generate actionable insights. The incident manager categorizes reported issues, prioritizes them based on urgency, and assigns them to appropriate waste authorities or collection teams. The route optimization module analyzes bin status and historical collection data to determine efficient waste collection paths, minimizing fuel consumption and operational delays. Notifications are automatically generated and sent to collection teams and administrative dashboards whenever bins reach critical levels or new incidents are registered. The system stores processed information within centralized databases for analytics, reporting, and future decision-making. Real-time updates are maintained using caching mechanisms to ensure instant synchronization between the backend server and client applications. Municipal administrators monitor activities through dashboards that provide analytics reports, live status updates, and operational control. Collection teams receive optimized routes and update task completion status, creating a feedback loop that continuously refines system performance. Overall, the data flow ensures seamless interaction between citizens, loT devices, authorities, and management applications to achieve efficient, automated, and. sustainable waste management operations.



Data Flow Explanation

1. Sensor detects bin level.
2. ESP32 sends data to cloud.
3. Cloud updates database.
4. Dashboard shows bin status.
5. If bin > 80%, notification sent.
6. Route optimization algorithm runs.

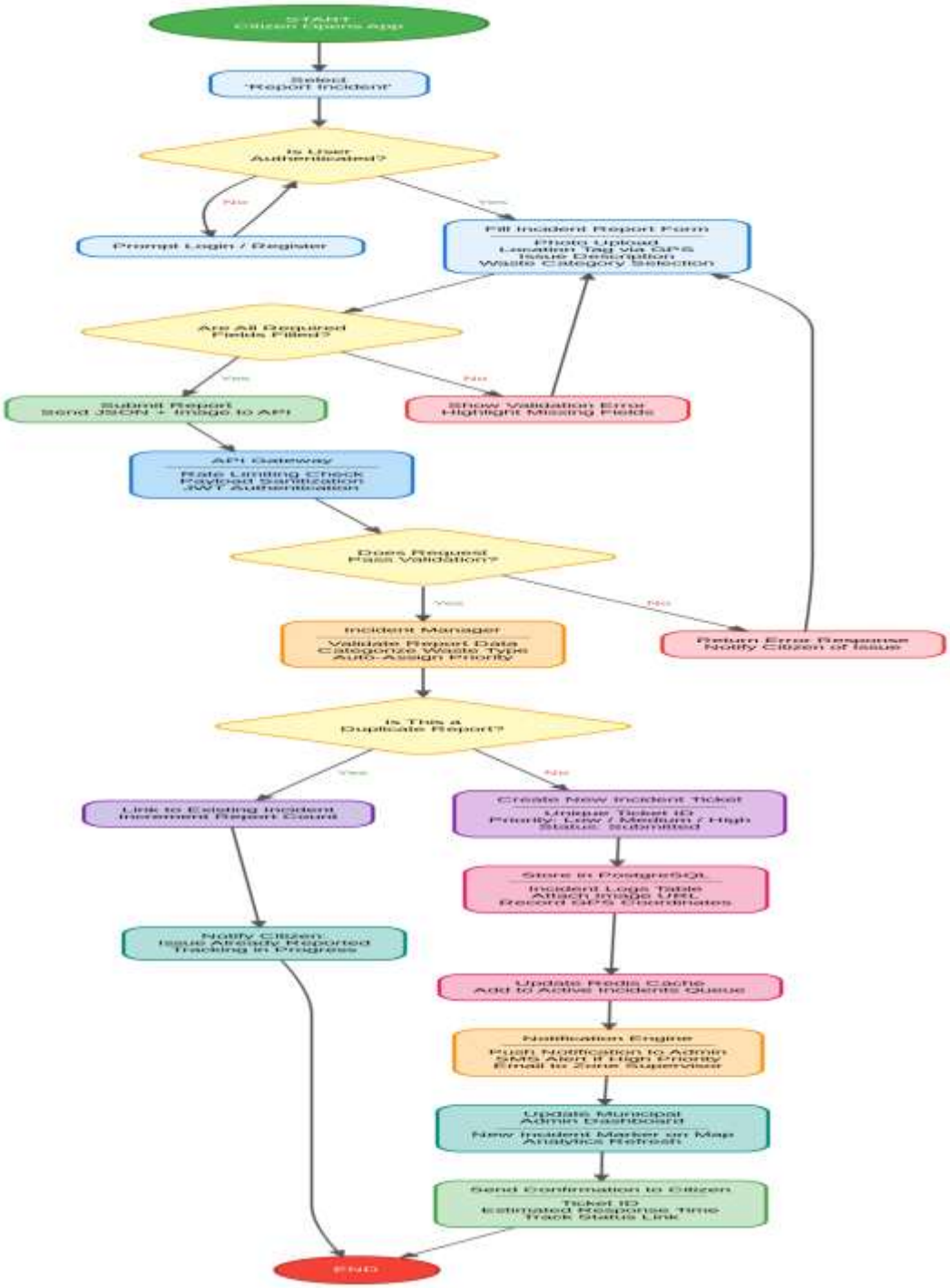
7. Driver receives optimized route.

8. After collection, driver updates status.

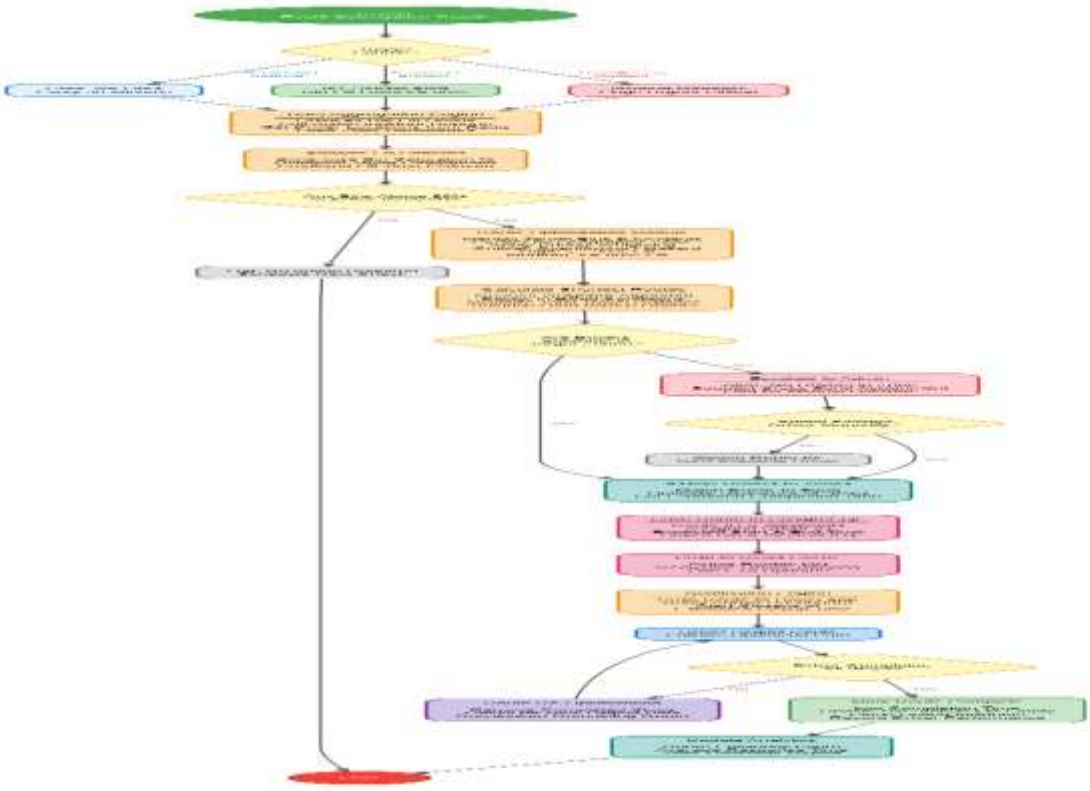
9. System marks bin as empty.

7. FLOWCHART LOGIC DESIGN

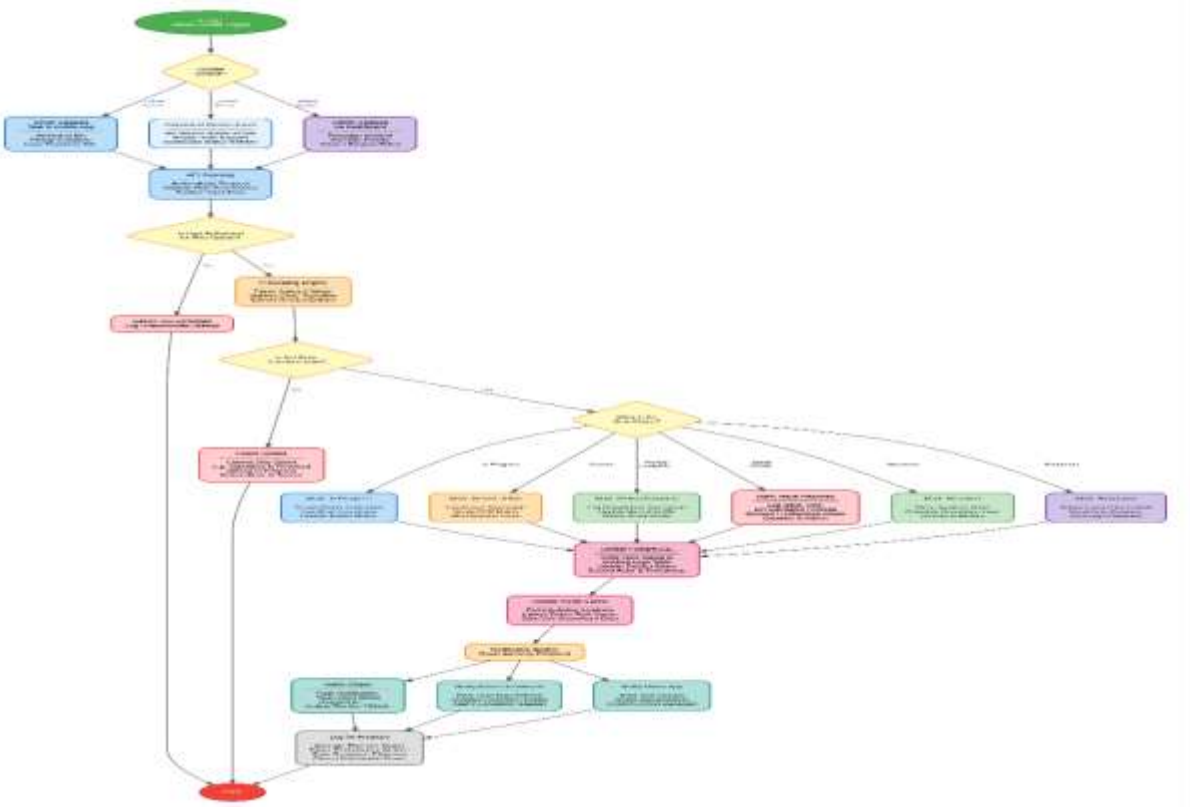
A) Report Incident Flow



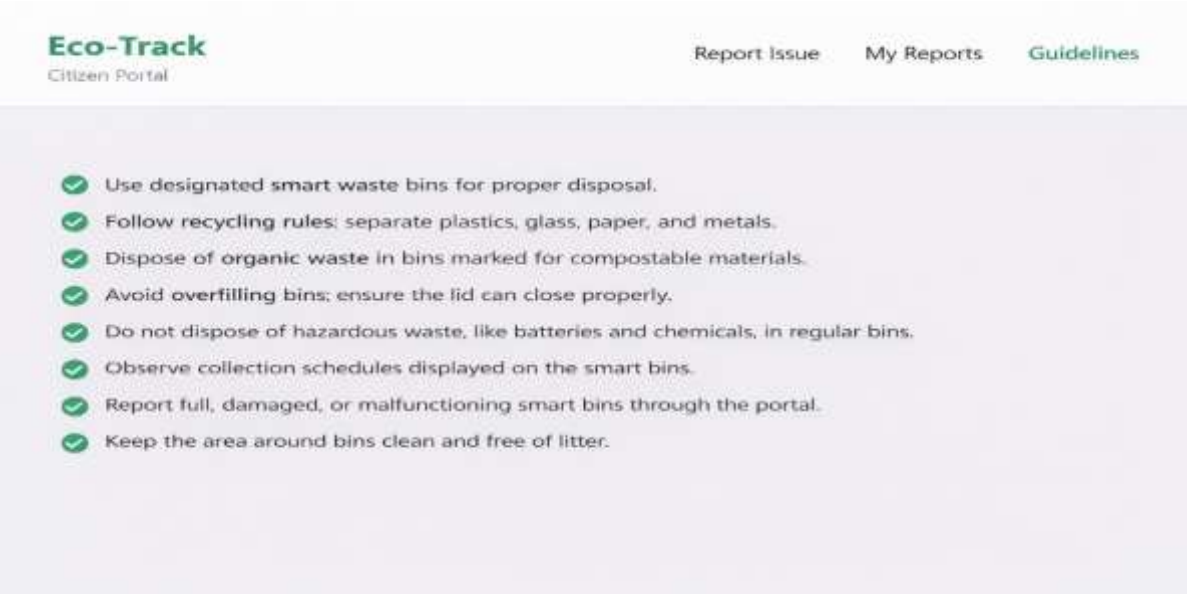
B) Route Optimization Flow

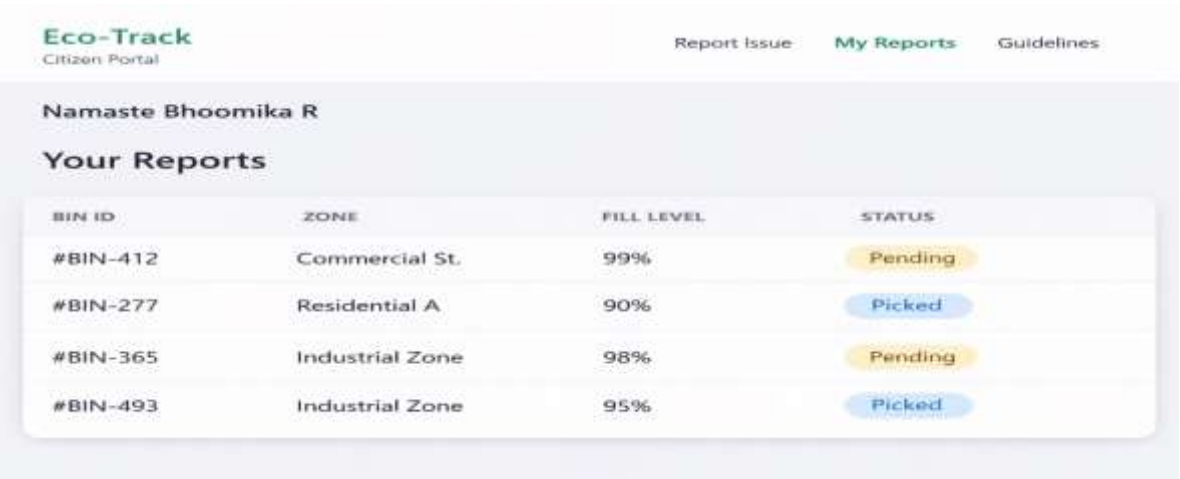


C) Status Update Flow

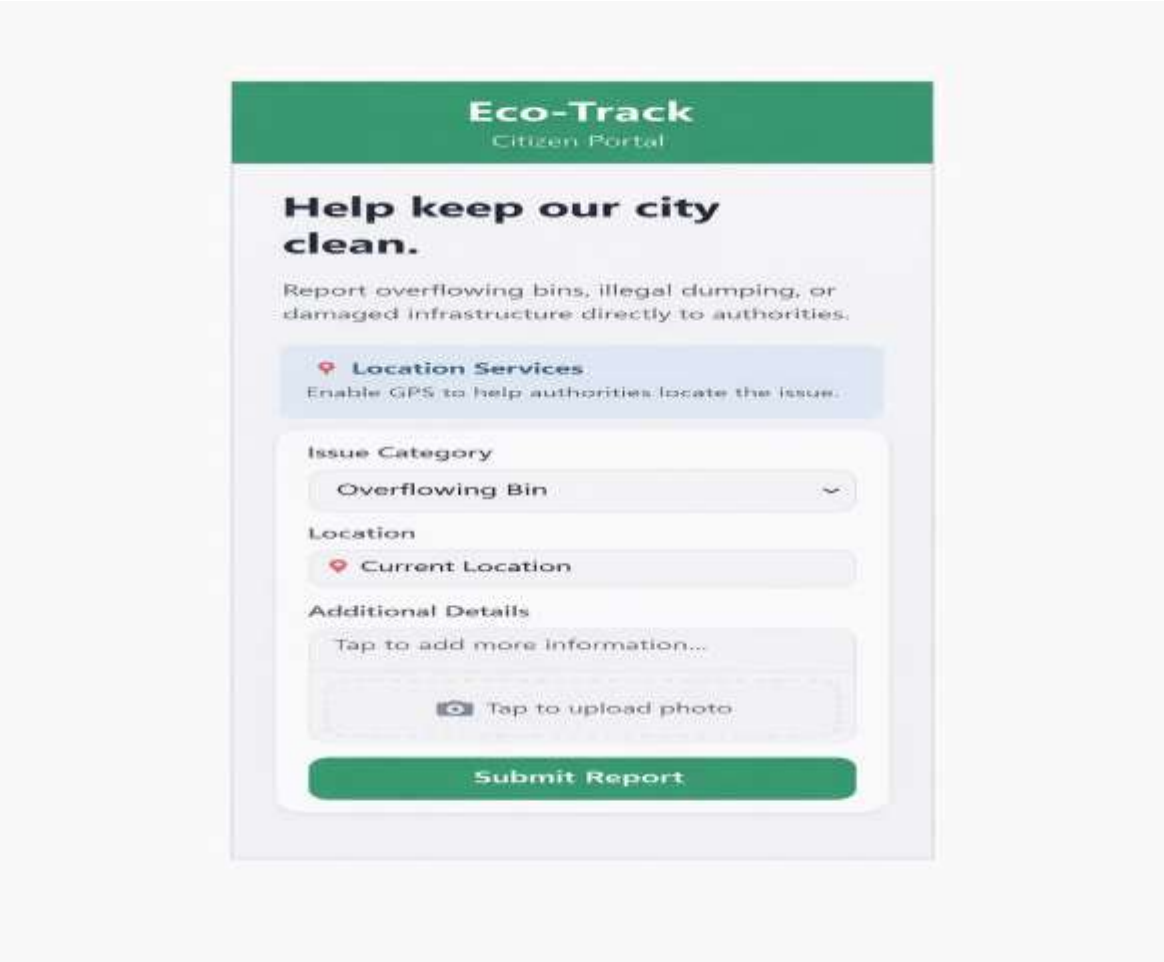


8. FIGMA TASK (UI DESIGN PLAN)





Eco-Track
Citizen Portal
Welcome Back!
Log in to your account
Email
Password
Forgot password?
Log In
Don't have an account? Sign Up


Eco-Track
Citizen Portal
Help keep our city clean.
Report overflowing bins, illegal dumping, or damaged infrastructure directly to authorities.
Location Services
Enable GPS to help authorities locate the issue.
Issue Category
Overflowing Bin
Location
Current Location
Additional Details
Tap to add more information...
Tap to upload photo
Submit Report

Citizen App Screens

1. Login/Register.
2. Report Waste.
3. Upload Photo.
4. Track Complaint Status.

Admin Dashboard Screens

1. Map view of bins.
2. Complaint list.
3. Bin status colour coding:
    - Green -Empty
    - Yellow -50%
    - Red -Full
4. Route assignment page.

9.HARDWARE IMPLEMENTATION PLAN

Components Required

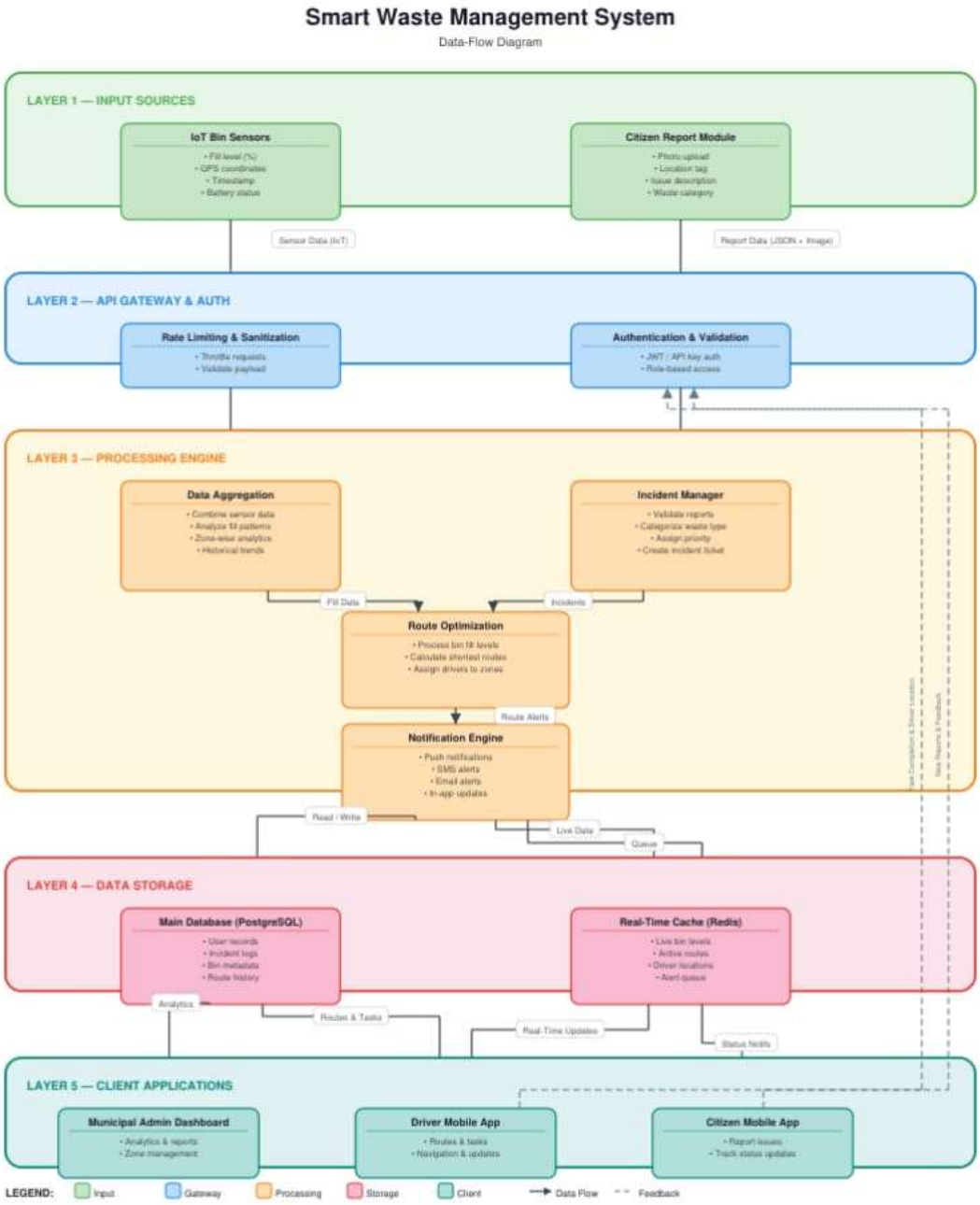
- ESP32.
- Ultrasonic Sensor (HC-SR04).
- Jumper wires.
- Breadboard.
- Power supply.

10.Working Logic

- Ultrasonic measures distance.
- Distance converted to fill percentage.
- If distance small-bin full.

- Data sent via Wi-Fi to cloud.

11.ROUTE OPTIMIZATION ALGORITHM



12. EXPECTED RESULTS

- Reduced overflow bins.
- Faster complaint resolution.
- 20-30% fuel saving.

- Real-time monitoring.
- Improved city cleanliness.

13. LIMITATIONS

- Internet dependency.
- Sensor damage risk.
- Initial setup cost.

14. FUTURE ENHANCEMENT

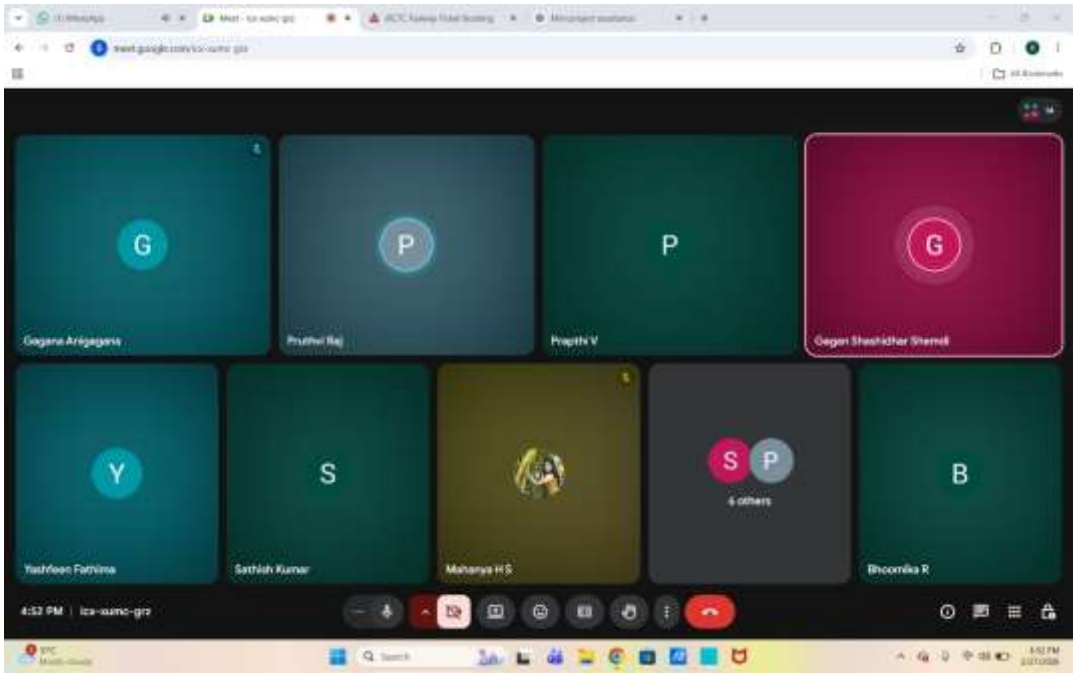
- AI-based waste prediction.
- Solar-powered bins.
- Waste segregation detection.
- Carbon emission tracking.

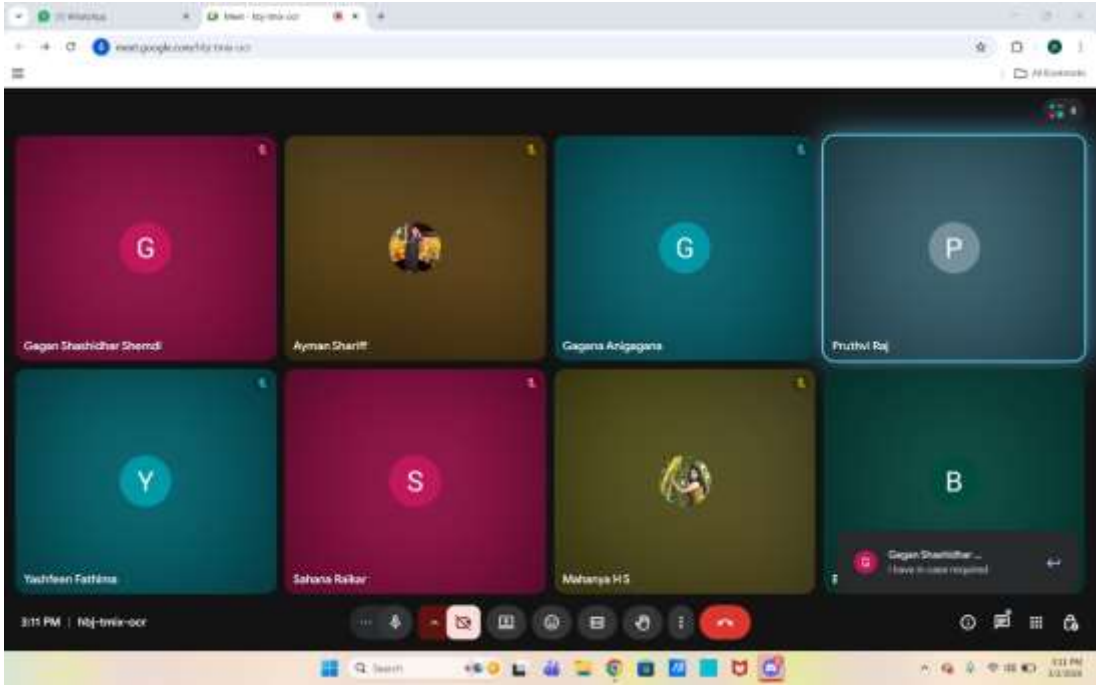
15. Conclusion

- Eco-Track: Smart Waste Management System Using IoT and Route Optimization is a practical and scalable solution designed to address the growing challenges of urban waste management. The system integrates IoT-enabled smart bins, cloud-based monitoring, citizen participation, and route optimization algorithms to create an efficient, data-driven waste collection process.
- By enabling real-time monitoring of garbage levels, Eco-Track helps prevent bin overflow and reduces environmental pollution. The inclusion of a citizen reporting feature bridges the communication gap between residents and municipal authorities, ensuring faster response to illegal dumping and sanitation issues. Additionally, the implementation of route optimization techniques minimizes fuel consumption, operational costs, and collection time, thereby improving overall efficiency.
- This project strongly aligns with **SDG 11: Sustainable Cities and Communities**, as it promotes cleaner urban environments, resource optimization, and smart city development. Through technological innovation and community involvement, Eco-Track contributes toward building sustainable, resilient, and environmentally responsible cities.

- In conclusion, Eco-Track demonstrates how IoT and intelligent algorithms can transform traditional waste management systems into smart, efficient, and sustainable urban solutions.

  These are the pictures of 3 official meetings conducted by the cohort{Thursday, Friday and Monday}.

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16. REFERENCE

1. Gemini Team and Google, “Gemini: A Family of Highly Capable Mul-timodal Models,” arXiv: Vol. 2312.11805v4 [Report], 2024. [Online]. Available: https://arxiv.org/abs/2312.11805v4

2. H. Dong and S. Xie, “Large Language Models (LLMs): Deployment, Tokenomics and Sustainability,” 2024, May 27. [Online]. Available: https://arxiv.org/abs/2405.17147

3. S. AlGhozali and S. Mukminatun, “Natural Language Processing of Gemini Artificial Intelligence Powered Chatbot,” Balangkas, vol. 1, no. 1, pp. 41–48, 2024. [Online]. Available: https://journal.amr-inc.org/

4. N. Rane, S. Choudhary, and J. Rane, “Gemini versus ChatGPT: ap-plications, performance, architecture, capabilities, and implementation,” Journal of Applied Artificial Intelligence, vol. 5, no. 1, pp. 69–93, 2024. [Online]. Available: https://doi.org/10.48185/jaai.v5i1.1052 CrossRef

5. L. A. Kadiyala, O. Mermer, D. J. Samuel, Y. Sermet, and I. Demir, “The Implementation of Multimodal Large Language Models for Hydrological Applications: A Comparative Study of GPT-4 Vision, Gemini, LLa Va, and Multimodal-GPT,” Hydrology, vol. 11, no. 9, p. 148, 2024. [Online]. Available: https://doi.org/10.3390/hydrology11090148 CrossRef

6. I.Nikolova-Stoupak, S. Bibauw, A. Dumont, F. Stas, P. Watrin, “LLM -Generated Contexts to Practice Specialised Vocabulary: Corpus Presentation and Comparison,” 35th Journées d'Etudes sur la Parole (JEP 2024) 31st Conference on Natural Language Processing (TALN 2024) 26th Meeting of Student Researchers in Computer Science for Natural Language Processing (RECITAL 2024), Toulouse, France, Jul. 2024, pp. 472–498. [Online]. Available: https://hal.archives-ouvertes.fr/hal-04623036

7. A.Bromfeld, Gemini, and ChatGPT, “Digital Revolution: AI Research Paper (By W. Isaacson),” 2024.

8. Y. A. Lee, “INVESTIGATING AI CHATBOT INTEGRATION IN ACADEMIC LIBRARIES: A CASE STUDY,” University of Oklahoma & School of Library and Information Studies, School of Library and Information Studies, 2024.

9. O. Herden and Cooperative State University Baden- Wuerttemberg, Germany, “Integration of Chatbots for Generating Code Into Introductory Programming Courses,” Journal-article, 2021.

10. N. P. Utami, F. R. Kembaren, and S. H. Daulay, “Exploring Higher Education Learners' Experience of Utilizing Gemini Chatbot in English Language Learning,” RETORIKA: Jurnal Ilmu Bahasa, vol. 10, no. 3, pp. 661–675, 2024. [Online]. Available: https://doi.org/10.55637/jr.10.3.10244.661-675

11. D. R. Ananda, M. Salmiah, and State Islamic University of North Sumatra, “Students' Perceptions on AI Technology: Gemini as a Writing Assistant Tool,” Linguistics and English Language Teaching Journal, vol. 12, no. 1, pp. 46–47, 2024.

12. “Sustainable Development Goals: Directive Principles For Sustainable India By 2030,” 2024. [Online].

13. “Sustainable Development- Issues and Perspectives,” 2024. [Online].

14. Google Cloud. (2024). Introduction to Vertex AI. https://cloud.google.com/vertex-ai/docs/start/introduction-unified-platform

15. Google Cloud. (2024). Forrester TEI of Vertex AI Report. https://cloud.google.com/resources/forrester-tei-of-vertex-ai-report?hl=en